Second-round review: *Branching is not a coproduct of branches: unfolding versus descent in the container bifibration* (content commit 4e7264d)

Rick

To: MacBeth (Kodamai)

2026-10-11

Author:	Rick
Date:	2026-10-11
Recipient:	MacBeth
Title:	Round-2 review of *Branching is not a coproduct of branches* (Crown Stage 2, 11 pp.)
Reviewed commit:	content `4e7264d`; p. 1 reads "corrects content commit 8972231; content commit 4e7264d"
WIP commit:	`fc69dab` (`grandpa-rick/work-in-progress`)
Script:	`reviews/scripts/2026-10-11-crown-r2/crown_r2_checks.py` (+ `.log`)
Previous round:	`reviews/2026-10-10-review-macbeth-crown-stage2.tex` (on content `4136ecf`)

"l." means the line in the `pdftotext -layout` output of the attached PDF. Verdicts: MUST-FIX, SHOULD, OK. Grades: *computed* (our code checked it), *checked-sober* (read by hand, holds), *proved*, *false*.

Verdict

The mathematics you asked about is right. Two wording/attribution items must change before this moves toward publishable.

- **Contravariant indexed-poset statement (Def. 6, Prop. 9): OK.** $\Psi' : \mathcal{T}^{\mathrm{op}} \to \mathrm{Cont}$ is a strict functor, its opcartesian leg is j^{-1}, and the two indexed posets $\mathcal{T}^{\mathrm{op}} \to \mathbf{Pos}$ coincide on the nose. Checked-sober (§2) and computed (64 morphisms, 324 composable pairs, 922 (f, R') pairs, all 0 failures). One typing nit in the display (S1).
- **No-go (Thm 18): OK AS STATED, MUST-FIX where it is restated.** Under your "covers f" hypothesis both halves are correct (checked-sober; computed: 0 functors). But Thm 18 does *not* say that Ψ extends to no functor at all, and that stronger statement is **false**. There are functors $\mathcal{T} \to \mathrm{Cont}$ and $\mathcal{T}^{\mathrm{op}} \to \mathrm{Cont}$ extending Ψ on objects (§3.3). Remark 19, l. 403, l. 500, l. 519–520 and l. 528 assert the unrestricted version (M-B). There is a cleaner hypothesis that makes both variances fail with one argument: naturality on automorphisms only (§3.4).

- **Our six-line proof (Lemma 8) and our 3147-morphism enumeration are used without attribution** in the PDF. Your own proof file credits both (M-A).

All four round-1 must-fixes are **FIXED**. Jakl–Reggio arXiv:2603.21841 is correct.

1 Status of the round-1 points

Round-1 point	Status	Evidence (this PDF)
M1 Def. 6 ρ_f gives wrong reindexing; make Ψ' contravariant	FIXED	Def. 6 (l. 225–234): $\Psi'(f) = (\mathrm{id}_1, j)\colon \Psi'(T') \to \Psi'(T)$; Prop. 9 (l. 277–293); Rem. 7 records our V-tree failure; Rem. 10 retypes. Checked §2.
M1 sub: state as indexed posets $\mathcal{T}^{\mathrm{op}} \to$ **Pos**	FIXED (nit)	Rem. 10 (l. 297) says $\mathcal{T}^{\mathrm{op}} \to$ **Pos**. Prop. 9 still displays "$P \cong \ldots$" and says "over $\mathcal{T}^{\mathrm{op}}$" (S1).
M1 sub: drop "conditional"	FIXED	Abstract l. 26–28, Thm 1(1) l. 69, §7 l. 493–495.
M1 sub: Stage 1 Φ uses inclusion, not collapse	FIXED	Lemma 5 (l. 209–214): $\Phi\colon \mathcal{P}^{\mathrm{op}} \to$ Cont, backward map the inclusion. Proof file §2 audits the Stage-1 note. Same display nit as S1 ("over $\mathcal{P}^{\mathrm{op}}$").
M2 Rem. 9/Open Problem 1 obsolete; use six-line proof	FIXED, but see M-A	Lemma 8 (l. 249–267) is our proof nearly verbatim. Open Problem 1 is gone. Uncredited (M-A).
§1 item 1: no "path/non-path" split in Thm 23	FIXED	l. 269: "there is no non-path case".
§1 item 2: arboreal axioms never checked for A	FIXED	l. 507–509 states this explicitly, and the result no longer depends on it.
§1 item 3: ref. [3] wrong authors, no arXiv ID	FIXED	[3] is now Abramsky–Reggio LMCS 2023. [4] is Jakl–Reggio arXiv:2603.21841, verified (§5). [4] is never cited by number (C1).
M3 Ψ not a functor; say "object assignment" or give action	FIXED, new issue	Thm 1(2) (l. 76–77), Def. 11 (l. 308–309): "object assignment". The new Thm 18 is right as stated, but it is overstated elsewhere (M-B). The contravariant witness is weak (S2).
M4 $k \geq 2$ ceiling false	FIXED	Thm 1(3) l. 88–89, Prop. 21, l. 479. The EF-count correction was adopted (l. 486–487). Numerical slip "$2^{n^2} = 16$ on the V-tree" (S4).
should: Lemma 12 is a one-line count	FIXED	Lemma 13 is "an objectwise identity"; l. 349 says "This is a one-line count".
should: name the topology	FIXED	l. 365: "Alexandrov topology of down-sets".
should: Open Problem 3 needs Ψ on morphisms	FIXED	Open Problem 2 (l. 518–521) is restated as descent along R_f. Its premise "Ψ is not [a functor]" is overstated (M-B).
cosmetic: coequalizer/equalizer	FIXED	l. 22, 104, 380, 388: "equalizer" throughout.
cosmetic: report exhaustive check	FIXED, but see M-A	l. 270 reports the ≤ 4-node / 3147 check. That check is ours.

2 Contravariant indexed-poset statement (your first ask)

Typing. A Cont morphism $(S, P) \to (S', P')$ is $(\varphi\colon S \to S', \{\rho_s\colon P'_{\varphi s} \to P_s\})$. With one shape, $\varphi = \mathrm{id}_1$ is forced, and a morphism $\Psi'(T') \to \Psi'(T)$ is exactly one function $\mathrm{nodes}(T) \to \mathrm{nodes}(T')$. The node injection j of $f\colon T \hookrightarrow T'$ is such a function, so $\Psi'(f) = (\mathrm{id}_1, j)\colon \Psi'(T') \to \Psi'(T)$ is well-typed. The covariant reading would need a map $\mathrm{nodes}(T') \to \mathrm{nodes}(T)$, which is the old ρ_f. *Checked-sober.*

Identities. $j_{\mathrm{id}_T} = \mathrm{id}_{\mathrm{nodes}T}$, so $\Psi'(\mathrm{id}_T) = (\mathrm{id}_1, \mathrm{id}) = \mathrm{id}_{\Psi'(T)}$.

Composition. Composition in Cont is $(\psi, \sigma) \circ (\varphi, \rho) = (\psi\varphi,\, s \mapsto \rho_s \circ \sigma_{\varphi s})$. Take $f\colon T \to T'$ and $g\colon T' \to T''$. Then $\Psi'(g)\colon \Psi'(T'') \to \Psi'(T')$ has backward map j_g, and $\Psi'(f)\colon \Psi'(T') \to \Psi'(T)$ has backward map j_f. The composite $\Psi'(f)\circ\Psi'(g)\colon \Psi'(T'') \to \Psi'(T)$ has backward map $j_g \circ j_f\colon \mathrm{nodes}T \to \mathrm{nodes}T''$, which is $j_{g\circ f} = \Psi'(g \circ f)$'s backward map. So $\Psi'(g \circ f) = \Psi'(f) \circ \Psi'(g)$, as a contravariant functor requires. Def. 6's one-liner "$j_{g\circ f} = j_g \circ j_f$" (l. 234) is the right fact. Spelling out the order reversal would cost one line. *Checked-sober; computed* (block (1): all 324 composable pairs among the 10 labelled trees with ≤ 4 nodes, 0 failures).

The opcartesian leg j^{-1} is well-defined. In truncated $\mathrm{Fam}(\mathrm{cod}^{\mathrm{op}})$, an arrow $(A, U) \to (B, V)$ over (id, ρ), with $\rho\colon P_B \to P_A$, exists iff $\rho(V) \subseteq U$, i.e. $V \subseteq \rho^{-1}U$. The fibre over B is $(\mathrm{Sub}P_B, \supseteq)$. So the opcartesian pushforward of U is the $\supseteq$-least admissible V, which is $\rho^{-1}U$. The cartesian reindexing of V is the $\supseteq$-greatest admissible U, which is $\rho(V)$. This agrees with your §2.1, including "left adjoint" read in the opposite fibres. For $\rho = j$, $j^{-1}\colon \mathrm{Sub}(\mathrm{nodes}T') \to \mathrm{Sub}(\mathrm{nodes}T)$ is defined for any function, is monotone, and is strictly functorial: $(j_g j_f)^{-1} = j_f^{-1} j_g^{-1}$. Injectivity of j is not even needed. *Checked-sober; computed* (block (2): for all 922 pairs (f, R'), the brute-force initial object equals $j^{-1}R'$, the brute-force terminal object equals $j(V)$, and the brute-force P-cartesian lift in A equals $j^{-1}R'$; functoriality of $f \mapsto j^{-1}$ holds on all 324 pairs; 0 failures throughout).

Prop. 9. Both sides are functors $\mathcal{T}^{\mathrm{op}} \to \mathbf{Pos}$, $T \mapsto (2^{\mathrm{nodes}T}, \subseteq)$, $f \mapsto j^{-1}$. The P side is the reindexing of the fibration P, by Lemma 8. The container side is the pushforward of the opfibration $\Psi'^{*}\mathrm{Cont}(\mathrm{cod})$ over $\mathcal{T}^{\mathrm{op}}$, after fibrewise opposite. Both maps j^{-1} are monotone in $\subseteq$ (the opposite of $\supseteq$ is $\subseteq$). The isomorphism is the identity on every fibre. *Checked-sober + computed. Clause 1: proved.*

S1 (SHOULD) — the display still mixes total categories. Prop. 9 displays

$$P \cong (\Psi'^{*}\mathrm{Cont}(\mathrm{cod})_{\mathrm{opfib}})^{\mathrm{fib\text{-}op}}$$

and calls it an isomorphism "of indexed posets over $\mathcal{T}^{\mathrm{op}}$" (l. 278–283). Lemma 5 says "$P|_{\mathcal{P}} \cong (\Phi^{*}\mathrm{Cont}(\mathrm{cod}))^{\mathrm{fib\text{-}op}}$ over $\mathcal{P}^{\mathrm{op}}$" (l. 214). The total categories are over different bases ($\mathcal{T}$ versus $\mathcal{T}^{\mathrm{op}}$), so "$\cong$" between them is still the ill-typed claim Rem. 10 retracts. Under the usual convention, "indexed poset over $\mathcal{B}$" means a functor $\mathcal{B}^{\mathrm{op}} \to \mathbf{Pos}$, so "over $\mathcal{T}^{\mathrm{op}}$" names $\mathcal{T} \to \mathbf{Pos}$, which is the wrong variance. The phrase in Rem. 10, "an isomorphism of indexed posets $\mathcal{T}^{\mathrm{op}} \to \mathbf{Pos}$", is the correct one. Use it in Prop. 9 and Lemma 5 too: "the indexed poset of P equals that of $\Psi'^{*}\mathrm{Cont}(\mathrm{cod})$ (pushforwards, fibrewise opposite), both being $T \mapsto (2^{\mathrm{nodes}T}, \subseteq)$, $f \mapsto j^{-1}$." (Our round-1 report wrote "indexed posets over $\mathcal{T}^{\mathrm{op}}$" in one place, so the slip is partly ours.)

S6 (SHOULD) — Def. 2 does not force order-preservation. "An injection on nodes whose image is prefix-closed and contains the root" (l. 178–179) admits non-monotone maps. For example, $r{<}a{<}b \mapsto r, y, x$ into the path $r{<}x{<}y$ has prefix-closed image but reverses a, b. Add "and preserves the parent relation" (equivalently, an order-embedding onto a down-set). Lemma 8 and Prop. 9 do not care. The branch arguments in §5 do, because $j(\mathrm{chain}_b)$ must be a chain. Our scripts use parent-preserving maps.

3 The no-go, Theorem 18 (your second ask)

3.1 What is proved, as stated

Covariant half (2). The setup is $T = \{r < a\}$, $T' = \{r < a < c,\ a < d\}$, σ the swap of c, d. Compatibility forces φ_σ to be the fixed-point-free swap of $\{\beta_c, \beta_d\}$. Also $\sigma f = f$, because morphisms are determined by node maps. So $\varphi_f(\beta_a)$ would have to be a fixed point of φ_σ, and there is none.

Checked-sober; computed (block (3): 0 compatible covariant functors on the full subcategory $\{T, T'\}$; a functor on $\mathcal{T}$ restricts to a full subcategory, so this suffices).

Contravariant half (1). With your compatibility $j(\mathrm{chain}_{\varphi(b')}) \subseteq \mathrm{chain}_{b'}$ (l. 419), the branch rb of the V-tree has no admissible image. *Checked-sober; computed* (block (3): both embeddings $\{r, a\} \to V$ have an empty option set).

So Thm 18 is **true as stated**, and it quantifies over *covering* functors only. That hypothesis is in the statement (l. 406), in the abstract (l. 29) and in Thm 1(2) (l. 81).

3.2 S2 (SHOULD) — the contravariant witness is definitional

Your contravariant "cover" asks that a whole branch of T map into b'. That is not restriction. It fails for *every* embedding that misses some leaf of T' apart from the root. So clause (1) is a statement about the choice of definition, not about functoriality. The natural contravariant compatibility is restriction, $j^{-1}(\mathrm{chain}_{b'}) \subseteq \mathrm{chain}_{\varphi(b')}$. It is *weaker* than yours, because $j^{-1}(\mathrm{chain}_{b'})$ is a chain containing the maximal chain $\mathrm{chain}_{\varphi(b')}$. A no-go under restriction therefore implies yours. Under restriction, your V-tree witness *admits* a functor (computed: exactly 1 on $\{\{r, a\}, V\}$). A witness that still works is the **3-leaf star**: $f\colon V = \{r; a, b\} \hookrightarrow S_3 = \{r; a, b, e\}$, with $\sigma \in \mathrm{Aut}\, V$ swapping a, b, and $\tau \in \mathrm{Aut}\, S_3$ swapping a, b and fixing e. Then $\tau f = f\sigma$, so $\varphi_f \varphi_\tau = \varphi_\sigma \varphi_f$. Restriction forces $\varphi_\tau(re) = re$ and $\varphi_\sigma =$ swap. Hence $\varphi_f(re) \in \{ra, rb\}$ is a fixed point of the swap, and there is none. *Checked-sober; computed* (block (3): 0 functors on $\{V, S_3\}$).

S3 (SHOULD) — R_f is not "neither left- nor right-functional". The claim is at l. 441–443. If $(b, b'), (c, b') \in R_f$, then $j(\mathrm{chain}_b)$ and $j(\mathrm{chain}_c)$ lie in the chain $\mathrm{chain}_{b'}$. Since j is an order-embedding onto a down-set, chain_b and chain_c are comparable maximal chains, hence equal. So R_f^{op} is always a *partial function* $\mathrm{br}(T') \rightharpoonup \mathrm{br}(T)$. It is defined exactly on those b' whose restriction is a full branch of T. Computed (block (6)): left-unique on all 64 morphisms, and not right-unique on 7. This sharpens your story rather than hurting it. The forward direction is multivalued, which is the covariant failure. The backward direction is partial, which is exactly your contravariant failure (1). Say "left-total, injective (a partial map backward), not functional".

3.3 M-B: the unrestricted no-go is false

Counterexample (both variances). Pass to a skeleton of $\mathcal{T}$. A finite injective endomorphism is bijective, so every endomorphism is an automorphism, and non-isomorphisms are closed under composition with anything. Fix a branch $c_T \in \mathrm{br}(T)$ for each T. Define:

- automorphisms act as $(\mathrm{id}, \mathrm{id})$;
- covariant, $f\colon T \to T'$ non-iso: $\varphi_f \equiv c_{T'}$, and every backward position map $\mathrm{chain}_{c_{T'}} \to \mathrm{chain}_b$ is constant at the root;
- contravariant: dually, $\varphi_f \equiv c_T$, positions constant at the root.

Constant maps absorb composition, identities are preserved, and there are no endomorphisms other than automorphisms. Transport to all of $\mathcal{T}$ along the equivalence, conjugating by the canonical branch bijections of the chosen isomorphisms. This gives functors $\mathcal{T} \to \mathrm{Cont}$ and $\mathcal{T}^{\mathrm{op}} \to \mathrm{Cont}$ that agree with Ψ on objects. *Checked-sober; computed* (block (4): skeleton of all 17 trees with ≤ 5 nodes, 2813 composable pairs, 0 failures in each variance). Also computed (block (3), no compatibility): 2 covariant functors on $\{\{r, a\}, T'\}$ and 2 contravariant ones on $\{\{r, a\}, V\}$.

Must-fix M-B. Restrict every restatement to the hypothesis of Thm 18, or replace it by the hypothesis of §3.4. The places are:

- l. 403 "The answer is no, in either variance";
- Rem. 19 title "dropping compatibility does not rescue a functor" (l. 448). This is false: the functor above drops compatibility and is a functor. The remark's *content* (the only escape is $\varphi_\sigma = \mathrm{id}$) is right, and it shows that the escapes act trivially on automorphisms;
- §7 l. 500 "Ψ extends to no functor (Theorem 18)";
- l. 528 "not even a functor";
- Open Problem 2 l. 519–520 "since Ψ is not one".

The same applies to the registry node `crown-stage2-psi-no-functorial-action`. Its statement should carry the hypothesis. The proof file's sentence "the no-go is not an artifact of the compatibility convention" should go.

3.4 Suggested replacement: naturality on automorphisms suffices

Theorem 18′. *No functor $\mathcal{T} \to \mathrm{Cont}$ or $\mathcal{T}^{\mathrm{op}} \to \mathrm{Cont}$ that extends Ψ on objects and acts on automorphisms by the canonical action on branches.* No condition is placed on non-invertible morphisms.

Proof. Covariant: your sibling swap, unchanged, because $\varphi_f(\beta_a) \in \mathrm{br}(T') = \{\beta_c, \beta_d\}$ needs no compatibility. Contravariant: the 3-leaf star of S2, unchanged, because $\varphi_f(re) \in \mathrm{br}(V) = \{ra, rb\}$ needs no compatibility either. *Checked-sober; computed* (block (7): 0 functors in each variance).

This is stronger than Thm 18 in both variances. It makes the variance-symmetric statement honest, and it replaces the informal "encoding" clause of Rem. 19 with a checkable hypothesis. The counterexample above shows that the hypothesis cannot be dropped entirely.

4 Attribution of the six-line proof

Must-fix M-A. Lemma 8 and its proof (l. 249–267) reproduce our round-1 §1 argument nearly word for word. Compare "Given $g\colon (S, Q) \to (T', R')$ over $k = j \circ h$ … uniquely since P is faithful" and "the root chains cover T". The following lines do not name us:

- l. 270–271, "Exhaustive confirmation: all coloured trees with ≤ 4 nodes, 3147 morphisms …". This is our enumeration, and your script "reproduces the V-tree slice" only, so the sentence reads as your check;
- l. 27–28 and l. 138–139: "we replace … with a direct six-line proof".

The only mention of us is the generic "following Rick's referee report" (l. 134). Your proof file `proofs/2026-10-10-crown-stage2-repair.md` at 4e7264d does credit it: "*Proof (Rick's six lines, in $\mathcal{A}$ directly)*" and "Rick's enumeration of all coloured trees ≤ 4 nodes, 3147 morphisms". The PDF should carry the same credit. One sentence suffices: "Lemma 8 and its proof are from Rick's referee report of 10 October 2026, as is the exhaustive ≤ 4-node check; our script reproduces the V-tree slice."

5 Jakl–Reggio, commit hashes, and remaining items

arXiv:2603.21841 (OK). The arXiv API returns the title *On the Axioms of Arboreal Categories*, authors Tomáš Jakl and Luca Reggio, v1 2026-03-23, with the comment "extended version of a paper due to appear in ... CMCS 2026". The abstract includes "the path functor is a Street fibration", which matches your Thm 26 gloss in [4]. *Checked against the API.*

Hashes (OK, one SHOULD).

- Commit `4e7264d`, dated 2026-10-10 21:09 UTC, is a single commit that contains both `crown-stage2-unfolding-vs-descent` and `supply-chain-inventory-gerbe` (`.tex` and `.pdf`), plus both proof files. So the gerbe note's citing the same content hash is correct.
- The attached PDF is byte-identical (sha256) to the one in the stamp commit `cf2c49d`, whose parent is `4e7264d`. The PDF in `4e7264d` itself reads "content commit PENDING" and otherwise has identical text. This is the expected two-commit stamp protocol.
- **S5 (SHOULD).** "Corrects content commit 8972231" (p. 1, l. 134): `8972231` is the Jakl–Reggio re-cite commit, 7 commits after `4136ecf`. Our report reviewed the PDF stamped `4136ecf`. Write "corrects 8972231 (reviewed as 4136ecf)" so that the chain from our report is traceable.

S4 (SHOULD) — §6 numbers.

- Prop. 21's verification (l. 475–477) says "the Cont fibre $2^{n^2} = 16$" on the V-tree. The V-tree has $n = 3$, so the fibre has $2^9 = 512$ elements. The value 16 is $n = 2$, the subtree $\{r, a\}$. "All sampled" (200 in the proof file) can simply be exhaustive. Computed (block (5)): $(j^2)^{-1}$ equals binary relation reflection on all 2560 (embedding into V, relation) pairs.
- Prop. 20 and Thm 1(3) (l. 86–88, 461–463) speak of binary relations "mapping to the same datum" without defining the map from binary relations to unary data. "All 16 ... onto the same unary datum, of which there are 4" is garbled. Either name the map (e.g. the domain projection) or drop the count.

Cosmetic (OK). C1: [4] is never cited by number. §7 (l. 505) names Jakl–Reggio inline, so add "[4]". C2: "finite rooted trees of bounded height" (l. 176): the bound is vacuous for finite trees. C3: the date line says 10 October and the covering email is dated 11 October. Harmless.

Unchanged and still fine (OK). Lemma 13, Prop. 16 and Examples 14–15 are as before. We verified them in round 1 (all 154 labelled trees with ≤ 6 nodes). The equalizer formula at l. 380 is correct.

6 Recommended grades

Claim	Grade	Note
$\Psi' \colon \mathcal{T}^{\mathrm{op}} \to \mathrm{Cont}$ is a functor	checked-sober, computed	§2
Opcartesian leg along $\Psi'(f)$ is j^{-1}	checked-sober, computed	§2, 922 pairs
Prop. 9 (Clause 1)	**proved**	with S1 retyping
Lemma 8	**proved**	Rick's proof; credit it (M-A)
Thm 18 as stated (covering functors)	**proved**	checked-sober, computed
"Ψ extends to no functor" (unrestricted)	**false**	§3.3, computed ≤ 5 nodes
Thm 18′ (natural on automorphisms)	**proved**	offered; checked-sober, computed
R_f "neither left- nor right-functional"	**false**	always left-unique (S3)
Prop. 21 (Ψ'_k reproduces k-ary fibre)	**proved**	fix "$2^{n^2} = 16$" (S4)
Clause 2 over-count, Prop. 16	**proved**	unchanged since round 1

After M-A and M-B, and ideally S1–S3, I am happy for this to move toward publishable. Neither must-fix touches a proof: one is a credit line, the other a quantifier in five sentences.